

Calculus II

Homework

Integration by parts

1. Let $x \in (0, 1)$. Express the following using x and $\sqrt{1 - x^2}$.

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|----------------------------|----------------------------|
| (a) $\sin(\arcsin(x))$. | (e) $\sin(2 \arccos(x))$. |
| (b) $\sin(2 \arcsin(x))$. | (f) $\sin(3 \arccos(x))$. |
| (c) $\sin(3 \arcsin(x))$. | (g) $\cos(2 \arcsin(x))$. |
| (d) $\sin(\arccos(x))$. | (h) $\cos(3 \arccos(x))$. |

2. Express as the following as an algebraic expression of x . In other words, “get rid” of the trigonometric and inverse trigonometric expressions.

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|--|------------------------------------|
| (a) $\cos^2(\arctan x)$. | (c) $\frac{1}{\cos(\arcsin x)}$. |
| (b) $-\sin^2(\operatorname{arccot} x)$. | (d) $-\frac{1}{\sin(\arccos x)}$. |

3. Rewrite as a rational function of t . This problem will be later used to derive the Euler substitutions (an important technique for integrating).

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|---------------------------|---|
| (a) $\cos(2 \arctan t)$. | (g) $\cos(2 \operatorname{arccot} t)$. |
| (b) $\sin(2 \arctan t)$. | (h) $\sin(2 \operatorname{arccot} t)$. |
| (c) $\tan(2 \arctan t)$. | (i) $\tan(2 \operatorname{arccot} t)$. |
| (d) $\cot(2 \arctan t)$. | (j) $\cot(2 \operatorname{arccot} t)$. |
| (e) $\csc(2 \arctan t)$. | (k) $\csc(2 \operatorname{arccot} t)$. |
| (f) $\sec(2 \arctan t)$. | (l) $\sec(2 \operatorname{arccot} t)$. |

4. Compute the derivative (derive the formula).

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|------------------------------------|---|
| (a) $(\arctan x)'$. | (d) $(\arccos x)'$. |
| (b) $(\operatorname{arccot} x)'$. | (e) Let arcsec denote the inverse of the secant function. Compute $(\operatorname{arcsec} x)'$. |
| (c) $(\arcsin x)'$. | |

5. (a) Let $a + b \neq k\pi$, $a \neq k\pi + \frac{\pi}{2}$ and $b \neq k\pi + \frac{\pi}{2}$ for any $k \in \mathbb{Z}$ (integers). Prove that

$$\frac{\tan a + \tan b}{1 - \tan a \tan b} = \tan(a + b) \quad .$$

(b) Let x and y be real. Prove that, for $xy \neq 1$, we have

$$\arctan x + \arctan y = \arctan \left(\frac{x + y}{1 - xy} \right)$$

if the left hand side lies between $(-\frac{\pi}{2}, \frac{\pi}{2})$.

6. Evaluate the indefinite integral. Illustrate the steps of your solutions.

$$(a) \int x \sin x dx.$$

$$(b) \int x e^{-x} dx.$$

$$(c) \int x^2 e^x dx.$$

$$(d) \int x \sin(-2x) dx.$$

$$(e) \int x^2 \cos(3x) dx.$$

$$(f) \int x^2 e^{-2x} dx.$$

$$(g) \int x \sin(2x) dx.$$

$$(h) \int x \cos(3x) dx.$$

$$(i) \int x^2 e^{2x} dx.$$

$$(j) \int x^3 e^x dx.$$

7. Evaluate the indefinite integral. Illustrate the steps of your solutions.

$$(a) \int x^2 \cos(2x) dx.$$

$$(b) \int x^2 e^{ax} dx, \text{ where } a \text{ is a constant.}$$

$$(c) \int x^2 e^{-ax} dx, \text{ where } a \text{ is a constant.}$$

$$(d) \int x^2 \frac{(e^{ax} + e^{-ax})^2}{4} dx, \text{ where } a \text{ is a constant.}$$

$$(e) \int \frac{1}{\cos^2 x} dx. \quad (\text{Hint: This problem does not require integration by parts. What is the derivative of } \tan x?)$$

$$(f) \int (\tan^2 x) dx. \quad (\text{Hint: This problem does not require integration by parts. We can use } \tan^2 x = \frac{1}{\cos^2 x} - 1 \text{ and the previous problem.})$$

$$(g) \int x \tan^2 x dx. \quad (\text{Hint: } \tan^2 x dx = d(F(x)), \text{ where } F(x) \text{ is the answer from the preceding problem.})$$

$$(h) \int e^{-\sqrt{x}} dx.$$

$$(i) \int \cos^2 x dx.$$

$$(j) \int \frac{x}{1+x^2} dx \quad (\text{Hint: use substitution rule, don't use integration by parts})$$

$$(k) \int (\arctan x) dx.$$

$$(l) \int (\arcsin x) dx.$$

$$(m) \int (\arcsin x)^2 dx. \quad (\text{Hint: Try substituting } x = \sin y.)$$

$$(n) \int \arctan\left(\frac{1}{x}\right) dx.$$

$$(o) \int \sin x e^x dx$$

$$(p) \int \cos x e^x dx$$

$$(q) \int \sin(\ln(x)) dx.$$

$$(r) \int \cos(\ln(x)) dx.$$

$$(s) \int \ln x dx$$

$$(t) \int x \ln x dx.$$

$$(u) \int \frac{\ln x}{\sqrt{x}} dx.$$

$$(v) \int (\ln x)^2 dx.$$

$$(w) \int (\ln x)^3 dx.$$

$$(x) \int x^2 \cos^2 x dx. \quad (\text{This problem is related to Problem ?? as } \cos x = \frac{e^{ix} + e^{-ix}}{2}).$$

8. Compute $\int x^n e^x dx$, where n is a non-negative integer.